Psychology is the scientific study of behavior and mental processes. *Behavior* includes all of our outward or overt actions and reactions, such as talking, facial expressions, and movement. The term *mental processes* refers to all the internal, covert (hidden) activity of our minds, such as thinking, feeling, and remembering. Why "scientific"? To study behavior and mental processes in both animals and humans, researchers must observe them. Whenever a human being observes anyone or anything, there's always a possibility that the observer will see only what he or she *expects* to see. Psychologists don't want to let these possible biases* cause them to make faulty observations. They want to be precise and to measure as carefully as they can—so they use a systematic** approach to study psychology scientifically.

1.6 The Scientific Approach

1.6 Recall the five steps of the scientific approach.

In psychology, researchers want to see only what is really there, not what their biases might lead them to see. They must apply all the principles of critical thinking, and this can be achieved best by using the **scientific approach**, an approach to research intending to reduce the likelihood of bias and error in the measurement of data.

behavior are description, explanation, prediction, and control. The scientific approach is a way to accomplish these goals of psychology.

- **Description: What Is Happening?** The first step in understanding anything is to describe it. *Description* involves observing a behavior and noting everything about it: what is happening, where it happens, to whom it happens, and under what circumstances it seems to happen.

  For example, a psychologist might wonder why so many computer scientists seem to be male. She makes further observations and notes that many "nontechies" stereotypically perceive the life and environment of a computer scientist as someone who lives and breathes at the computer and surrounds himself with computer games, junk food, and science-fiction gadgets—characteristics that add up to a very masculine ambiance.

  That's what *seems* to be happening. The psychologist's observations are a starting place for the next goal: Why do females seem to avoid going into this environment?

- **Explanation: Why Is It Happening?** Based on her observations, the psychologist might try to come up with a tentative explanation, such as "women feel they do not belong in such stereotypically masculine surroundings." In other words, she is trying to understand or find an *explanation* for the lower proportion of women in this field. Finding explanations for behavior is a very important step in the process of forming theories of behavior. A **theory** is a general explanation of a set of observations or facts. The goal of description provides the observations, and the goal of explanation helps build the theory.

  The preceding example comes from a real experiment conducted by psychologist Sapna Cheryan and colleagues (Cheryan et al., 2009). Professor Cheryan (who teaches psychology at the University of Washington in Seattle) set up four experiments with more than 250 female and male student participants who were not studying computer science. In the first experiment, students came into a small classroom that had one of two sets of objects: either Star Trek® posters, video-game boxes, and Coke™ cans, or nature posters, art, a dictionary, and coffee mugs (among other things). Told to ignore the objects because they were sharing the room with another class, the students spent several minutes in the classroom. While still sitting in the classroom, they were asked to fill out a questionnaire asking about their attitude toward computer science. While the attitudes of male students were not different between the two environments, women exposed to the stereotypically masculine setup were less interested in computer science than those who were exposed to the nonstereotypical environment. The three other similar experiments yielded the same results. Later studies found that when women were exposed to role models who dressed and acted according to the computer science stereotyped image, those women showed decreased interest in computer science as a career as well as decreased expectation of success in that field (Cheryan et al., 2011; Cheryan et al., 2013). In two similar follow-up studies with high school students, the researchers found that providing adolescent girls with an educational environment that did not fit current computer science stereotypes seemed to increase their interest in computer science courses (Master et al., 2015).

- **Prediction: When Will It Happen Again?** Determining what will happen in the future is a *prediction*. In the original Cheryan et al., study, the prediction is clear: If we want more women to go into computer science, we must do something to change either the environment or the perception of the environment typically associated with this field. This is the purpose of the last of the four goals of psychology: changing or modifying behavior.



Fancy/Alamy Stock Photo

Is this an environment that you would want to work in? Some researchers have wondered if your answer might be influenced by gender.

theory
a general explanation of a set of observations or facts.

- **Control: How Can It Be Changed?** The focus of control, or the modification of some behavior, is to change a behavior from an undesirable one (such as women avoiding a certain academic major) to a desirable one (such as more equality in career choices). Professor Cheryan suggests that changing the image of computer science may help increase the number of women choosing to go into this field.

Not all psychological investigations will try to meet all four of these goals. In some cases, the main focus might be on description and prediction, as it would be for a personality theorist who wants to know what people are like (description) and what they might do in certain situations (prediction). Some psychologists are interested in both description and explanation, as is the case with experimental psychologists who design research to find explanations for observed (described) behavior. Therapists may be more interested in controlling or influencing behavior and mental processes, although the other three goals would be important in achieving this objective.